Baron Eiley

Baron.eiley@gmail.com || (213) 399 – 4456 || linkedin.com/in/baron-eiley-5655a8196

Professional Experience

Lockheed Martin Space - ASIC and FPGA Design Engineer **March 2023 – Present**

FPGA RTL Design and Architecture

- Developed mission sensitive, bare-metal RTL for long-range high-altitude vehicles, for arm-and-disarm of critical hardware functions. Designs included custom encoding schemes, robust finite state machines, and logic to read in real-time sensor data.
- Designed Simulink-based models to generate synthesizable RTL for signal processing modules including interpolators, decimators, and cascaded integrator-comb (CIC) filters.
- Integrated auto-generated RTL into top-level FPGA architectures for Software Defined Radio (SDR) platforms and validated top-level hardware modules through automated CI/CD verification pipelines.
- Validated signed fixed-point RTL for arithmetic operations that represent accelerometer data to confirm FPGA mathematical integration to produce velocity and distance calculations for high altitude interceptor vehicles.
- Architected FPGA-based top-level modules that instantiate algorithms that interface with System-on-Chip (SoC) hard hard-processors for AMD ZYNQ and Ultrascale+ devices utilizing high-performance bus protocols such as AXI and APB.
- Integrated Vitis Model Composer image processing algorithms, including Complementary Median Filters and Logarithmic Matched Filters, RTL top-level architectures for FPGA proof-of-concept demonstrations.

Lockheed Martin Space - Electro-Optical Engineer **August 2022 – March 2023**

C# UX/UI Firmware Development

- Developed custom scripting language using C# to command-and-control special read out integrated circuits for highly sensitive Geiger Mode Avalanche Photo-Diode sensors.
- Designed UX/UI GUIs in C# to interface with test hardware associated with Hardware in the Loop (HWIL) and Standardized Test Equipment (STEs).
- Wrote C# code to utilize dynamic link libraries to interface National Instruments test equipment to be controlled and monitored via custom graphical user interfaces.

Graduate Projects

RapidLaneFPGA: FPGA DPU Based Highway Lane Segmentation System

- Developed a MobileNetV2 lane-segmentation CNN model and deployed it to FPGA using AMD Vitis AI and PyTorch. Project utilized Deep Processing Unit (DPU) hardware on 224×224 images.
- Achieved 77.10% DICE accuracy on FPGA vs. 77.8% software baseline with INT8 quantization at 10 FPS (~100ms latency) on AMD ZCU104 Ultrascale+ device, demonstrating near-real-time inference.

FPGA Based LeNET Neural Network for Digit Recognition

- Implemented LeNet-based MNIST digit classifier using Vitis High Level Synthesis and deployed FPGA.
- Hardware output was validated against software golden data and validated FPGA inference on PYNQ-Z1, successfully classifying digits 0–9.

Swimmers Pose Estimation Using Computer Vision and AI/ML

- Developed a computer vision pipeline for swimmer pose estimation using MediaPipe object detection, Segment Anything Model, and YOLOv7 to isolate swimmers in image and generated coordinates to train on a custom neural network.

FPGA Based Flight Controller, Search and Rescue Drone

- Developed FPGA-based flight controller for a six-propeller search-and-rescue drone. The drone hosted Zynq-7000 FPGA/SoC chip to process RC inputs and generate PWM motor control signals within the FPGA fabric.
- Designed a power distribution board for onboard camera and lighting systems.

Technical Skills

Vivado & Vitis High Level Synthesis (HLS), Verilog, SystemVerilog, VHDL, C++ Hardware Design
MATLAB/SIMULINK, Signal Processing, Control Systems Modeling, and Hardware Arithmetic Validation
QuestaSim, FPGA Hardware Simulation & Validation
Software Programming Languages, Working knowledge of Python and C/C++
Microsoft Excel, Word, & PowerPoint

Education

Master of Science in Electrical Engineering **December 2025**
San José State University, GPA: 3.44 San Jose, CA
Emphasis: Digital System Design and Artificial Intelligence

Bachelor of Science in Electronic Systems Engineering **May 2022**
California State Polytechnic University, Pomona, GPA: 3.49 Pomona, CA
Emphasis: Digital System Design, Robotics, and Control

Awards

Science Spectrum Trailblazer @ Becoming Everything You Are (BEYA) **March 2025**

- Awarded the Science Spectrum Trailblazer distinction at the 39th BEYA STEM Global Competitive Conference